

BEACON ONCOLOGY REFERENCE LABORATORY (SYNTHETIC FACILITY)
450 Synthetic Technology Parkway, Cambridge, MA 02100 | CLIA ID: 00D8874123 | Director: Dr. Arthur Sterling, MD, FCAP (Synthetic)

Liquid Biopsy Colorectal ctDNA Early Screening Report

Circulating Tumor DNA Methylation & Somatic Mutation Targeted Panel (Synthetic Demonstration)

Patient Demographics

Name: Harold K. Sample
DOB: 04/18/1964
Sex: Male
Patient ID: SYN-MRN-5529184

Ordering Clinician

Provider: Dr. Sophia Alvarez
Facility: New England Cancer Center
NPI: 9457896321
Location: Boston, MA

Specimen Details

Specimen ID: SYN-SPEC-CRC-7731
Collection Date: Jul 28, 2026
Received Date: Jul 29, 2026
Report Date: Aug 4, 2026
Specimen Type: Blood / Plasma

Result: Positive (Elevated Risk for Colorectal Malignancy)

Status: Final
Panel: Colorectal ctDNA Multi-Marker

Test Result Summary: Elevated circulating tumor DNA (ctDNA) markers detected in synthetic sample. Aberrant SEPT9 promoter hypermethylation and somatic mutations in KRAS and TP53 were identified in plasma cfDNA.

Quantitative and Qualitative Biomarker Findings

Target Marker	LOINC	Result	Reference Range	Interpretation
SEPT9 Methylation	77983-5	Positive	Negative	Abnormal (Positive)
KRAS Mutation Analysis	48018-6	p.G12D (VAF: 1.8%)	Not Detected (< 0.05% VAF)	Abnormal
BRAF Mutation Analysis	48018-6	Not Detected	Not Detected (< 0.05% VAF)	Normal
TP53 Mutation Analysis	48018-6	p.R273H (VAF: 2.3%)	Not Detected (< 0.05% VAF)	Abnormal
Total ctDNA Fraction	Local	2.1%	< 0.1%	Abnormal

Clinical Interpretation

The presence of circulating tumor DNA with SEPT9 promoter methylation and concurrent KRAS p.G12D and TP53 p.R273H variants strongly indicates the presence of shed colorectal neoplastic DNA. Diagnostic colonoscopy with high-definition mucosal inspection and radiological abdominal evaluation is urgently recommended. Clinical correlation with patient symptoms and family history is required.

Methodology & Limitations

Methodology: Plasma ctDNA undergoes dual-chemistry extraction. Methylation of Septin 9 is quantified using real-time PCR with fluorescent hybridization probes. Somatic variant profiling across hotspot exons of KRAS, BRAF, and TP53 is performed using ultra-deep amplicon NGS (minimum 20,000x coverage depth) on Illumina NextSeq.

Limitations: False-positive methylation or mutations may rarely occur due to clonal hematopoiesis of indeterminate potential (CHIP). Liquid biopsy results should be verified by endoscopic visualization and tissue histology.